

Report:

This project is a simple loan calculator written in C++ with Qt GUI. The calculator can calculate the monthly payments, loan amount, interest rate, number of payments, and loan balance based on user inputs. The GUI allows easier interaction, while the command line interface provides flexibility for quick calculations.

As this code is from 2009 so it has a lots of bugs, messy code and in old style.

So here we add some checks for validation
code:

```
// Validation/checks

if (calculator.getAmount() <= 0) {

    std::cout << "Error! Loan amount can't be negative\n";

    return 1;

}

if (calculator.getInterest() <= 0) {

    std::cout << "Error! Interest rate can't be negative\n";

    return 1;

}

if (calculator.getPeriodTotal() <= 0) {

    std::cout << "Error! Total loan period must be positive\n";

    return 1;

}

if (calculator.getPeriodElapsed() < 0) {

    std::cout << "Error! Elapsed period cant be negative\n";

    return 1;

}
```

- We Use long double instead of float and powl() instead of pow() which helps the program handle big numbers correctly and avoid mistakes in calculations. This makes the loan calculator more accurate and safe to use.

i.e:

```
float LoanCalculator::calculateNumberPayments() --> long double
LoanCalculator::calculateNumberPayments()
```

- We make a config file to keep default values (like interest or fees) separate from the code. This way, we can change these values easily without editing the program itself.
- Git commit history

```
C:\Users\Hp\legacy-calc-2009>git log --oneline
ce333e9 (HEAD -> assignment2, origin/assignment2) Final submission: organized repo and added report
973f3ba Added Utils and config files
801942d Updated README and other changes
847e865 (origin/master, origin/HEAD, master) Added compilation information
d9ff0427 Nicer formatting on Readme and added better app help text
71f0456 Added readme and additional app help text
c3e0801 Added a Qt Gui and scons compilation. The makefile was generated by Qt.
0e10569 Changes to use libCmdLineParser
12ee333 Removed compileCommands.txt
4d09e8b Made methods inline and removed some compiler errors
d99e446 First version
```